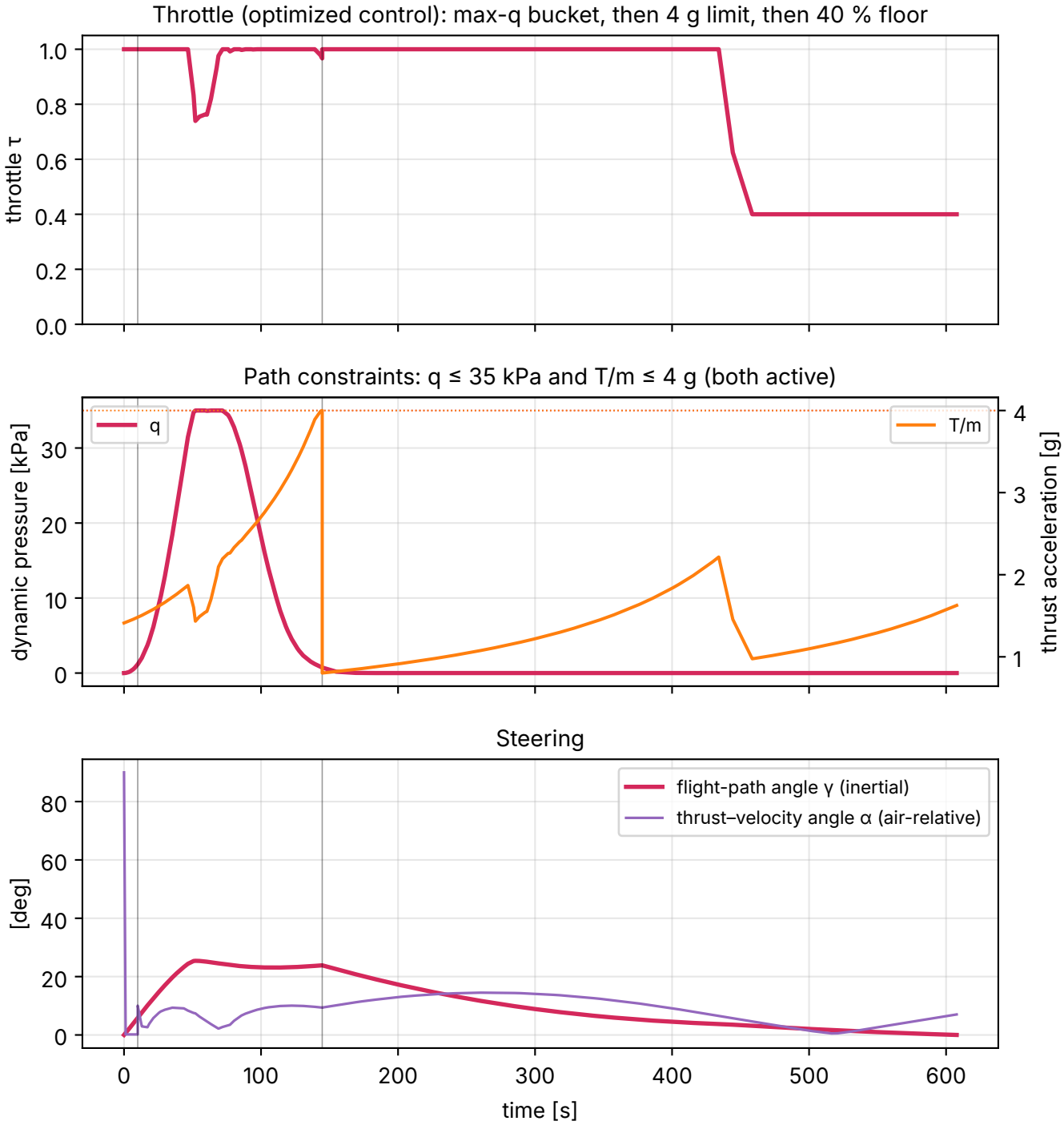
Throttle (optimized control): max-q bucket, then 4 g limit, then 40 % floor
throttle τ
1.0
0.8
0.6
0.4
0.2
0.0
Path constraints: q ≤ 35 kPa and T/m ≤ 4 g (both active)
dynamic pressure [kPa]
thrust acceleration [g]
q
T/m
30
20
10
0
4
3
2
1
Steering
[deg]
flight-path angle γ (inertial)
thrust–velocity angle α (air-relative)
80
60
40
20
0
0
100
200
300
400
500
600
time [s]